

Master's Thesis

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The Effect of Ketamine on Protein Synthesis and Neural Oscillations in the
Hippocampus

Introduction

Modern neuroscientists understand the brain as a network of interconnected neurons, itself composed of subnetworks with distinct functions. The role of these networks is the transmission and processing of sensory information and internal biological signals, which are integrated to generate behavior and maintain homeostasis. Information is in the form of electrical impulses called action potentials that are transmitted through neurons. These specialized cells of the nervous system have arbor-like processes called dendrites that sprout from the cell body, which are covered in tiny protuberances called dendritic spines. A second kind of membrane process called axons forms connections with the dendritic spines of other neurons. Through these connections called synapses, electrical signals carrying information can be relayed to other neurons that form part of a network. The brain uses this system to generate highly specialized computations that shape behavior and, presumably, the internal life of the mind. I am interested in how pushing this system to adopt certain uncommon configurations, in our case via pharmacological interventions thought to mimic and treat psychiatric pathologies, can give us insight into the biological nature of mental

disorders. Subsequently I will further elaborate the conceptual framework we use to describe this system, which is implemented to share the experiments and interpretations of their results presented in this thesis.

The transmission of information from one neuron to another is complex and utilizes a varied collection of chemical neurotransmitters to transmit the signal across the synaptic cleft that exists between the pre and postsynaptic cells. There are two main kinds of neurotransmission. One corresponds to signals that increase the likelihood of postsynaptic firing, referred to as excitatory neurotransmission, which are primarily mediated by the release and uptake of glutamate (Zhou & Danbolt, 2014). The second corresponds to signals that decrease the likelihood of postsynaptic firing, referred to as inhibitory neurotransmission, which are mediated by gamma-aminobutyric acid (GABA). One of the crucial functions of inhibition is to synchronize network activity and prevent runaway excitation (Avoli & Krnjević, 2016). Excitatory neurotransmission is necessary for signal propagation across the network, but without inhibitory neurotransmission the network could easily enter states of excitotoxicity and cease performing functional computations (Avoli & Krnjević, 2016; Buzsáki & Wang, 2012). It is crucial to maintain an excitation/inhibition (E/I) balance in the network for it to be able to process information effectively (Zhou & Yu, 2018). Too much excitatory drive and network activity becomes hyper-synchronous or epileptic, too much inhibition and network activity becomes uncoordinated before shutting down, which can lead to schizophrenia (Buzsáki & Wang, 2012; Singer, 1999; Zhou & Yu, 2018).

A neuronal network can shift its E/I balance across time, making its activity more or less synchronous. Shifts that are too extreme can lead to aberrant brain states

diagnosed as mental disorders (Zhou & Yu, 2018). In the work I will be presenting, I study how shifts in neural activity due to ketamine are related to changes in cellular metabolism, specifically protein synthesis. I use the framework presented thus far to link the biological insight from our experiments to our understanding of the biological nature of aberrant brain states associated with mental disorders. Our work is part of an overarching project whose goal is to bridge different levels of biology, from the behavioral and psychological to the physiological and molecular, in order to understand the neurobiology of cognition. In sections 2 and 3 I will present original data concerning the physiological and molecular effects of ketamine. For the remainder of this introduction and section 1, I will finish presenting the conceptual framework and summarize the scientific literature concerning the study of ketamine, respectively, in order to contextualize our contributions.

Subnetworks in the brain are composed of interconnected neurons that come in a variety of cell types, yet they can also be differentiated based on the neurotransmitter they release, i.e. glutamatergic principal cells (PCs) and GABAergic inhibitory cells (INs). The categorization of neurons into types is highly complex and can be done so on the base of function (excitatory vs. inhibitory), as well as genetically (based on unique genes expressed), physiologically (based on patterns of firing), morphologically (based on shape of dendrites and axons), and anatomically (based on position occupied in brain structures). For example, although all INs release GABA which shunts the propagation of action potentials, they encompass a wide variety of genetic types (not limited to PV, SST or VIP-expressing cells), physiological types (slow vs. fast-spiking) and morphological/anatomical types (basket cells vs. interneurons with long-

projections). Some cell types have determined features across categories. For example, PV-expressing interneurons are also fast-spiking and envelop targeted neurons with their neurites forming a basket-like structure (Buzsáki, 2001).

Different cell types compose functional subnetworks within a brain region, and different brain regions will make use of distinct assemblies of cell types organized anatomically in specialized configurations to execute distinct forms of information processing. One brain region that is known for its highly organized anatomical structure and segregation of different cell types is the hippocampus, which has been associated with information processing related to cognitive functions like learning, memory and spatial navigation as well as relating environmental cues to each other (Sweatt, 2004). The hippocampus is a crucial brain structure that has been heavily studied when it comes to mental disorders and their pathobiology, namely the correlation between antidepressant treatments and adult neurogenesis in the dentate gyrus (DG) and altered gamma oscillations in the hippocampus of patients diagnosed with schizophrenia (Kang et al., 2016; Malberg et al., 2000; Uhlhaas et al., 2008). This makes the hippocampus an excellent brain structure to study physiological and molecular changes associated with cognitive function and mental disorders.

In our work, we take wild-type BL-6 mice as model organism to study the effects of drugs which influence plasticity (such as ketamine) on the hippocampus. Neural activity in the brain can be recorded by inserting small electrodes into the region of interest and 'listening' to the fluctuations in membrane potential for the patch of tissue the electrode is adjacent to. One can measure individual action potentials when sampling the electrical signal at high rates (>10kHz) and singling out individual neuron

spiking, but one can likewise listen to larger oscillations associated with collective network activity at slower frequencies, called the local field potential, or LFP. We can get more insight into the state of information processing of a brain region by looking at the dynamics of the LFP instead of the activity of single cells. To construct a robust population-level signal we use the LFP of a recording electrode, i.e. the extracellular fluctuations of voltage in the vicinity of neuronal membranes. The LFP can be spectrally decomposed and partitioned into separate frequency bands (delta: 1-4Hz, theta: 4-8Hz, alpha: 8-12Hz, beta: 12-30Hz, gamma 30-140Hz), which have been linked to different physiological functions. For example, theta oscillations are known to be linked to active whisking and running in mice and organizes neural firing across the whole hippocampus according to its troughs and rises (Buzsáki & Draguhn, 2004; Kleinfeld et al., 2016).

Gamma oscillations, which encompass a large range in frequency space can be split into slow (30-50Hz) and fast (65-140Hz) gamma, are associated with more local computations performed by hippocampus subnetworks. For example, slow gamma oscillations from the *Cornu Ammonis* 1 (CA1) subfield have been found to cohere in phase with activity from the CA3, yet fast gamma oscillations in CA1 cohere with activity from the medial entorhinal cortex (MEC) (Colgin et al., 2009). Information processing in the hippocampus is shaped in distinct ways by these two circuits. Information from CA3 to CA1, transmitted through the Schaffer collaterals to the *stratum radiatum* (SR) of CA1, is shaped by the internal structure of the hippocampus established by previous experiences. Meanwhile information from MEC to CA1, from layer III of MEC to the *stratum lacunosum-moleculare* (SLM) of CA1, is shaped more by real-time sensory

stimuli. Thus, each kind of gamma implies a different kind of local computation in the context of the hippocampus.

High frequency fluctuations of population-level activity are intimately tied to the activity of fast-spiking inhibitory neurons (Bartos et al., 2002, 2007). Theoreticians have modeled gamma oscillations from excitatory-inhibitory (E-I) and inhibitory-inhibitory (I-I) interactions, and in both cases the fast-spiking rate of inhibitory activity is key to generate gamma oscillations (Buzsáki & Wang, 2012). The relationship between E-I and I-I connections and gamma oscillations has led researchers to consider changes in gamma oscillations to be shaped by molecular changes in INs or inhibitory synapses and *vice versa* (Ainsworth et al., 2016; Lee et al., 2009).

Synaptic changes that strengthen or weaken the connections between neurons are defined as synaptic plasticity and can happen in many different ways that often occur in tandem. Dendritic spines can grow or shrink by adding more phospholipids into their membrane, new receptors can be inserted into or removed from the active zone of the synapse, receptors can be phosphorylated to increase or decrease their binding efficiency, the amount of second messenger proteins can increase or decrease as to alter the sensitivity of the postsynaptic signaling machinery, and so on (Citri & Malenka, 2008; Kandel et al., 2014). Changing the strength of synapses at the molecular level means changing the strength of E-I and I-I connections, which can alter network activity patterns. Slight changes could be thought of as necessary for tweaking network activity according to environmental demands. Too drastic of a change, or a suboptimal configuration of synaptic plasticity due to, for example, genetic variations, would throw off the E/I balance in the network. The resulting E/I imbalance would thus affect the

capacity of the network to perform crucial computations, leading to cognitive or behavioral impairments.

Gamma oscillations are known to be altered in mental disorders such as schizophrenia. Researchers have shown, for example, that in schizophrenic patients there is a decrease in the degree of entrainment of gamma oscillations to 40Hz stimuli, which is observed in healthy brains (Uhlhaas et al., 2008). Pharmacological studies where NMDA receptor antagonists such as phencyclidine (PCP) and ketamine have been found to mimic symptoms of psychosis. Furthermore, post-mortem studies have found that schizophrenic brains have a deficit of N-methyl D-Aspartate (NMDA) receptors, a crucial component in the synaptic plasticity machinery that controls calcium influx into neurons essential for the control of protein synthesis (Davis & Squire, 1984; Nakazawa & Sapkota, 2020; Newcomer, 1999). Together, these findings have led to the hypothesis that there is an NMDA receptor hypofunction in mental disorders like schizophrenia, which could then cause population-level activity to be less organized by external stimuli.

The evidence of gamma changes and NMDA receptor hypofunction is consistent with the idea our framework offers: systematic molecular deficits at the points where the neuronal network relays information between cells could be altering the physiological signals of the entire population, i.e. disruption of normal gamma oscillations. Interestingly, it has also been found that at low concentrations, NMDA receptor antagonists such as ketamine but not PCP (which has a higher binding affinity to the NMDA receptor) are also able to decrease symptoms related to major depression and even treatment resistant depression (Krystal et al., 2019; Singh et al., 2016; Zarate et

al., 2006). This implies something interesting: for a healthy brain, pharmacologically inducing NMDA hypofunction mimics psychosis, yet for a brain suffering from depression the same treatment can result in relieving depressive symptoms and returning to a healthier state. Inhibition of glutamatergic neurotransmission in a healthy brain can trigger psychosis, but some inhibition at the right spot in a depressed brain can relieve depressive symptoms. This makes us recall the concept of E/I balance in subnetworks across brain regions. Given the role of fast-spiking PV-expressing INs in the production of gamma oscillations, perhaps the deficit of NMDA receptors in schizophrenia mimicked by NMDA receptor antagonists like ketamine impacts INs in a disproportioned manner, thus altering plasticity at crucial E-I and I-I synapses.

NMDA receptors are one of many receptors present at synapses that influence plasticity. Besides its function as an ionic channel able to conduct current across the dendritic membrane and depolarize the cell, it also controls *de novo* protein synthesis, an essential metabolic process that plays a central role in synaptic plasticity. Signaling other than from the NMDA receptors can influence *de novo* protein synthesis, but no matter the source of the control signal, the rate of protein synthesis will shape how synaptic connections are able to remodel themselves by determining the availability of raw materials: synaptic proteins. By measuring relative amounts of newly synthesized proteins after ketamine, an NMDA receptor antagonist, we can gain insight into how a metabolic process essential for synaptic plasticity is implicated in changes related to gamma oscillations, psychosis-like behavior, and the reduction of depressive symptom. In the work presented here I hypothesize that altered gamma oscillations in the hippocampus due to ketamine treatment are in part due to synaptic plasticity in INs that

requires an increase of *de novo* protein synthesis mediated by ketamine. I will subsequently present our initial data that has given shape to this hypothesis but requires further experimentation to comprehensively validate.

By targeting different levels of neurobiology, we can offer a more integrated conception of how psychosis or pharmacological treatments of depression operate under a single framework that involves the tuning of synaptic connections of subnetworks composing brain regions. We give special importance to inhibitory connections in the hippocampus, but it is likely that similar changes must occur across brain regions during psychiatric treatments. Our overarching project offers an integrative conception of the biology behind mental disorders, particularly that of schizophrenia as modeled by the psychotomimetic effects of ketamine in wild-type mice. In doing so it presents the opportunity to develop diagnostic and treatment strategies that target the underlying biology of the system rather than just the psychiatric symptoms.

Section 1

Ketamine was first synthesized in response to a demand for effective anesthetics that could minimize the adverse cognitive and behavioral side effects of barbiturates. In the early 20th century, the drug phencyclidine (PCP) was first synthesized by Victor Maddox as a new kind of anesthetic, yet it notably caused behavioral side effects that mimicked psychosis. Ketamine was developed by Calvin Stevens during his time working for Parke-Davis Laboratories as a derivative compound of PCP, and used in the first human clinical trials as an anesthetic by Guenter Corssen and Edward Domino.

The new drug required higher doses to induce the same psychotomimetic effects as PCP while still retaining similar anesthetic properties. Ketamine was able to induce a dissociative state that worked as an analgesic without necessarily causing psychotic symptoms as severe as PCP. It was coined by its discoverers as a “dissociative anesthetic” and was considered an advance for midcentury surgical practices (Domino et al., 1965; Corssen & Domino, 1966; Li & Vlisides, 2016). Although ketamine was approved by the FDA for clinical use in the 70’s (as Ketalar), its biological mechanism of action was a mystery. In the following decades research led by David Lodge found that both PCP and ketamine worked by blocking NMDA receptors, which led to the hypothesis that aberrant glutamatergic neurotransmission caused psychotic-like symptoms (Anis et al., 1983; Lodge & Mercier, 2015).

From the 50’s to the 90’s the field of neuroscience began integrating ideas from the study of nerve physiology, molecular biology and behavioral psychology. Models of biological learning emerged that saw changes in synaptic strength mediated by particular molecular channels to be the basic biological underpinnings of memory formation and learning. A central molecular player in this field was the NMDA receptor, which had a crucial role in determining whether signals being transmitted between neurons should reinforce or weaken their synaptic connections (Citri & Malenka, 2008; Kandel et al., 2014). Naturally, ketamine and other NMDA receptor antagonists became subject of much interest, especially given the already rich history of their recorded psychotomimetic properties. The scientific interest in the role of the NMDA receptor in glutamatergic transmission and the plasticity of neuronal networks, as well as the pharmacological evidence that NMDA receptor antagonism induced psychotic-like

symptoms drew the attention of clinicians back to ketamine, but this time from the field of psychiatry.

New frameworks emerged that attempted to explain cognitive function from a biological point of view. For example, in the context of memory, disrupting neural activity in hippocampal circuits dramatically impacted certain learned behaviors (Sweatt, 2004). Conceptual advancements in neuroscience in the 80's and 90's pushed psychiatrists to rethink the way they understood schizophrenia. The connection between NMDA receptors, excitatory neurotransmission and psychosis-like symptoms led to investigations on how excitatory and inhibitory interactions at specific regions of the brain shaped brain function and could lead to psychosis-like symptoms. In the early 2000's researchers at Yale University working on the psychotomimetic effects of NMDA receptor antagonists found that ketamine, as opposed to other known antagonists with psychotomimetic properties, could work as an antidepressant at low doses (Berman et al., 2000; Krystal et al., 2019; Zarate et al., 2006). This unprecedented discovery further supported the idea that mental disorders could be explained at the biological level by disruptions in the balance of excitation and inhibition in neuronal networks, something which could be manipulated for therapeutic ends through careful pharmacological intervention.

In the subsequent decades ketamine received special attention due to its distinctive dose dependent effects that had high clinical importance. Chemically, ketamine is very similar to PCP. It consists of a cyclohexanone ring that has a methylamino group and a 2-chlorophenyl group attached to carbon at position 2 in the ring. These two groups at position 2 can attach themselves in two different ways to the

cyclohexanone ring, producing two distinct enantiomers: R- and S-ketamine. Several research groups have found distinct effects for the different enantiomers. Structural modeling has shown that the different chiral structures produce different binding affinities to the NMDA receptor: S-ketamine binds more strongly than R-ketamine (Zhang et al., 2021). Studies in rodents also showed that while S-ketamine tends to have stronger psychoactive effects, R-ketamine still works as an effective antidepressant (Yang et al., 2015; Zanos & Gould, 2018).

One of the remarkable qualities of ketamine is that it has both rapid and robust antidepressant effect that can last for weeks. Ketamine contains polar groups that give it hydrophilic properties, as well as aromatic carbons that give it lipophilic properties, enabling it to easily pass the blood brain barrier and effectively interact with lipid-rich brain tissue (Peltoniemi et al., 2016). Ketamine is metabolized rather quickly. It has a half-life of 2-3 hours in adult humans and its main metabolite is norketamine. Although ketamine is rapidly metabolized, the drug can influence brain function for weeks (Singh et al., 2016; Zanos & Gould, 2018; Zarate et al., 2006). The process of ketamine metabolism is complex and outside the scope of this review, yet it is worth mentioning that some of its byproducts like (2R,6R)-hydroxynorketamine (HNK) have been found to also exert similar antidepressant effects as ketamine, complicating the story concerning ketamine's antidepressant mechanism of action (Zanos et al., 2016; Zanos & Gould, 2018).

Ketamine has been found to primarily target NMDA receptors due to its affinity to their open channel domain. NMDA receptors have a magnesium ion blocking the channel pore during rest. When the surrounding membrane is depolarized by an

excitatory event, the magnesium ion becomes repelled from the pore leaving the channel open. Additionally, in order for the NMDA receptor channel to be able to conduct a current, the excitatory event must involve the presynaptic release of glutamate. The binding of glutamate to the extracellular domain of the receptor induces conformational changes in the channel protein that allows ions to flow through and conduct current in the form of excitatory postsynaptic potentials (EPSPs) or serve as second messengers in molecular cascades that shape neuronal function. Ketamine blocks this process by preventing the conduction of ions through the NMDA receptor channel pore when it becomes open (Kavalali & Monteggia, 2020; Zanos & Gould, 2018; Zhang et al., 2021). Abstracting the molecular biology a bit, we can think of ketamine as a signal jammer that is only activated whenever a message is trying to be relayed between a pre and postsynaptic neuron using the NMDA receptor. Pathological brain states, like psychosis, are characterized by disorganized neurophysiology, that is, an imbalance between neural excitatory and inhibitory signaling. Ketamine's capacity to randomly jam NMDA receptor dependent neural signaling across the network during its presence in the tissue works as a first pass to understand the rapid psychotomimetic effects it can have at high doses.

NMDA receptors are a family of receptors that differ in their expression patterns across neuronal subtypes and cellular compartments (Monyer et al., 1994; Paoletti et al., 2013). Ketamine does not target all NMDA receptor subtypes equally (Khlestova et al., 2016; Zanos & Gould, 2018). Different subtypes have different affinities to ketamine, although all are inhibited by getting blocked at their open pore region (Zhang et al., 2021). NMDA receptors are composed of four subunits. Each subunit is a distinct

protein with 7 transmembrane domains, four of these subunits assemble into a tetramer structure with a central pore region that opens and closes. Every NMDA receptor has two N1 subunits, while the remaining two subunits can be N2(A-D). The most common NMDA receptors are those containing the N2A and N2B subunits. While N2A containing NMDA receptors tend to be located at the active zone of the synapse, the N2B containing kind inhabits more peripheral zones of dendritic spines (Paoletti et al., 2013). Ketamine has a higher binding affinity to N2A than N2B containing NMDA receptors due to the differences in amino groups exposed in the pore region that ketamine interacts with (Zhang et al., 2021). Interestingly, researchers have found that using N2A or N2B specific inhibitors produce distinct behavioral effects, where N2A specific inhibition has been more associated with psychotic-like behavior and N2B specific inhibition seems to be tied to antidepressant effects (Gerhard et al., 2020; Ibrahim et al., 2012; Park et al., 2024; Preskorn et al., 2008; Zanos & Gould, 2018). In recent years it has been found that ketamine's effects are not purely mediated by NMDA receptors. Ketamine can also act through pathways involving α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptors and opioid receptors (Gupta et al., 2011; Levinstein et al., 2023; Zaytseva et al., 2023). Research on ketamine's ability to affect other receptors in neuronal membranes is young, yet these findings have elucidated that ketamine is a drug with multiple binding partners and complex interactions.

One of the ways the field has begun addressing the question of ketamine's clinically relevant effects is by turning its focus beyond its pharmacology and understanding how it impacts cellular and physiological processes that mediate synaptic and neural network function. While molecular neurobiologists have focused on how

ketamine's direct interactions with NMDA, AMPA and other receptors impact neuron's ability to undergo synaptic plasticity via the modulation of pathways involving *de novo* protein synthesis, receptor trafficking and spine growth, neurophysiologists have delved into the way ketamine might be preferentially targeting specific neuronal subtypes, like GABAergic interneurons, that strongly shape neural oscillations and network function in different brain regions (Autry et al., 2011; Gerhard et al., 2020; Li et al., 2010; Nakazawa & Sapkota, 2020; Zanos & Gould, 2018).

Ketamine's neurobiological effects have been researched primarily in the hippocampus and the medial prefrontal cortex (mPFC). Depression is known to cause tissue atrophy in the hippocampus, and antidepressants have been found to counteract this by triggering neurogenesis in the DG (Kang et al., 2016; Malberg et al., 2000). The mPFC is also highly implicated in cognitive functions like decision making and emotional processing (Simon et al., 2021). Atrophy in prefrontal regions affects aspects of executive functioning and personality, which are heavily implicated in the symptomatology of mental disorders like schizophrenia and depression (Chai et al., 2011; Pizzagalli & Roberts, 2022). Recent studies have begun venturing beyond these two regions and targeting ketamine's effects in regions like the lateral habenula, known to mediate reward learning mechanisms (Hikosaka, 2010; Yang et al., 2018), and the role that connections between the hippocampus and mPFC play in the dynamic time course of ketamine's effects (Carreno et al., 2016; Zanos & Gould, 2018). Systemic administration of ketamine at high doses causes anesthesia while lower doses have cognitive effects similar to those associated with damage to regions like the hippocampus or the mPFC. Rapid 'jamming' of neurotransmission by ketamine in these

regions is thought to be the basis for ketamine's rapid antidepressant effects (Kavalali & Monteggia, 2015). Meanwhile the impact on the cellular processes upstream from the rapid process of synaptic transmission are thought to mediate the long-lasting effects by physically changing the connectivity of the neuronal networks in these regions (Kim et al., 2023).

Section 2

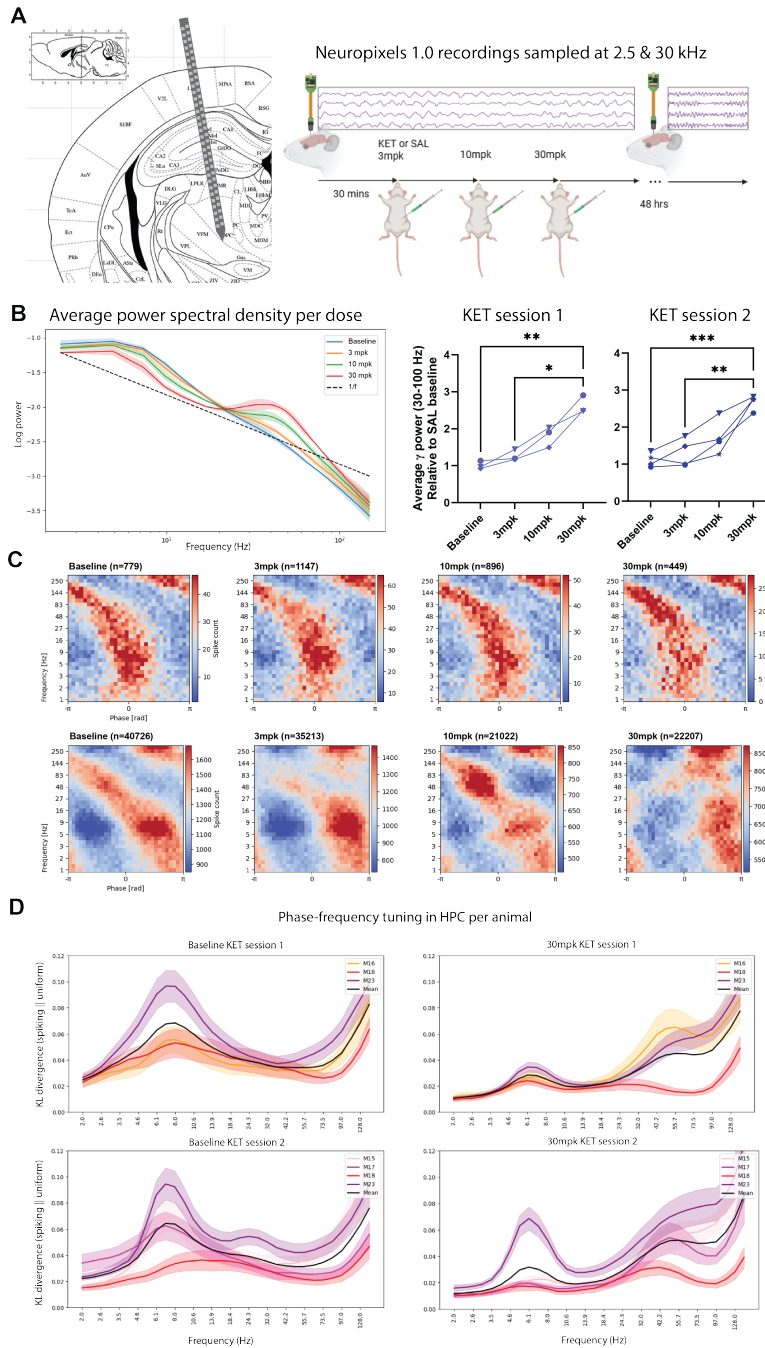
In this section we will discuss the effects of ketamine on gamma oscillations in the the hippocampus and the methods, experiments and analyses we performed to study this phenomenon. The hippocampus is a brain region known for its highly organized layered structure that creates a well defined circuit across its different subfields. The Fenton lab has many years of experience trying to understanding the function of the hippocampus and its role in cognition. We use silicone probes with high electrode density to record hippocampal LFPs. With these signals we can study the activity of specific neurons, population dynamics, and how population dynamics relate to neural firing. We used this approach to probe the changes in neural activity, specifically gamma oscillations and how neural firing is organized by gamma during and after ketamine treatment to understand what changes may underlie ketamine's acute and long-lasting effects.

In our experiments we use the Neuropixels 1.0 linear probes, which consist of a 3.9mm long and 60 μ m thick silicon shank that has 196 pairs of 20 μ m x 20 μ m recording electrodes arranged along its length. We inserted the probe in the dorsal hippocampus

(1.24-1.25mm lateral, -1.95 posterior to bregma, tip 3.6-3.8mm below dura, 10 degrees angle from midline) and recorded from the bottom-most third of the electrodes, which spans across the CA1 and DG as well as some of the motor cortex above and thalamus below (Figure 1A, left). Our recordings were each 2 hours long and done on head fixed mice placed on a small arena suspended on a cushion of air that enabled mice to move their feet, simulating locomotion (NeuroTar system). This experimental setup allowed us to perform stable extracellular recordings of neural activity in the hippocampus during acute ketamine treatment and gain insight into how individual neurons and population-level dynamics were altered in their electrophysiology due to ketamine.

Our ketamine treatment paradigm was designed to emulate a standard treatment paradigm in a clinical setting. Mice that underwent treatment were given three doses of saline-diluted ketamine of increasing doses (3mg/kg, 10mg/kg, and 30mg/kg) administered intraperitoneally (IP), each 30 minutes apart (Figure 1A, right). The first 30 minutes of the 2 hour session was used to determine baseline neural activity. In order to study persistent effects of ketamine, the same animals underwent a second full treatment two days later. The data was processed using Kilosort2 in order to extract the single unit spikes and a standard signal processing pipeline used in the lab to get the LFPs across the entire span of the recording electrodes and their corresponding current source density (CSD).

We first investigated ketamine's acute effects on the fluctuations of extracellular potentials across the hippocampus. Our approach consisted of decomposing the CSD into its different frequencies using a fast Fourier transformation (FFT) or a wavelet transformation and quantifying the relative strength or amplitude of each frequency



component in the signal across ketamine doses. Oscillatory activity at specific frequency ranges are known to be physiologically significant and their relative strength within the LFP follows a power law (Buzsáki & Draguhn, 2004; Singer, 1999). We primarily focused our analysis on the theta (4-8 Hz) and gamma (30-100Hz) ranges. These two frequency ranges are specially associated with the organization of neuronal firing patterns in the hippocampus. Particularly, theta is known to establish a kind of gating modulation that potentiates neuronal excitability during its peaks and suppresses firing during

its troughs, while gamma oscillations are related to the activity of fast-spiking inhibitory neurons, like PV interneurons, which are known to locally organize neural spiking (Bartos et al., 2002; Buzsáki & Draguhn, 2004; Buzsáki & Wang, 2012).

We found that during ketamine the power of gamma oscillations in the hippocampus is potentiated relative to the rest of the frequency spectrum, while theta oscillations were suppressed during ketamine, and that this effect was exaggerated at higher doses (Figure 1B). A decrease in theta power and increase in gamma can be interpreted as an introduction of interference in the normal theta signal and an increase in the coordination of fast-spiking inhibitory neurons. Next we checked how the activity of individual spiking units identified from the same extracellular recordings changed their tuning to gamma and theta oscillations due to ketamine.

To isolate the individual spiking units we used the kilosort2 algorithm, which is able to identify whether a spike came from the same putative neuron using the spike waveform, timing and location relative to the recording electrode. We used the collection of identified spiking units and their corresponding spikes to estimate the degree to which their activity was organized by gamma or theta oscillations. What we mean by this is whether the spiking activity of a putative neuron happens at random instances across the different phases of the oscillation of interest (not organized by that frequency band), or it occurs consistently at specific phases of the oscillation (activity is organized by the oscillation). For each putative neuron identified we constructed a phase-frequency tuning histogram with 32 bins corresponding to 32 evenly spaced phases of an oscillation and 36 frequency bins. We computed how many spikes fell on each of these 2-dimensional bins to then analyze its cell-specific distribution. We observed that when

we computed the phase-frequency tuning histogram for all cells in a recording, there were some units that showed changes in their gamma-phase tuning due to ketamine (Figure 1C). This suggests that there are specific cells being impacted by ketamine, and that it is modifying the degree to which they are organized by gamma oscillations. We found that these cells had high firing rates suggesting that PV interneurons could be increasing their tuning to gamma oscillations.

For each animal we recorded from, we computed a tuning distribution across frequencies. For each cell in the recording of a given animal we calculated the Kullback-Leibler (KL) divergence between the distribution of spikes across phase bins and a uniform distribution, which tells us how non-random is the firing pattern of a cell for a given frequency. The resulting KL divergence scores were averaged across all cells in a given recording to get a distribution of cell tuning across frequencies. From this analysis we were able to see that in fact, cells across animals were tuned to frequencies pertaining to theta and gamma band oscillations (Figure 1D). Furthermore, when spiking was partitioned into time windows corresponding to the different ketamine doses, we observed that ketamine increased gamma tuning at higher doses (Figure 1, C and D). Because we had recordings for the same animals across two sessions that were two days apart, we were also able to compare how the baseline phase-frequency tuning was affected by ketamine in a long-lasting manner. We observed that when we compared the tuning from the baseline on the first day to the tuning of the baseline on the second day (two days after ketamine), there was one animal that demonstrated a sustained increase in its gamma tuning (Figure 1D, bottom). This result points to the possibility of a potential physiological mechanism by which ketamine has its long-lasting

effects: a persistent change in gamma tuning. Yet our results at this time can only suggest such a case, not confirm it. In the next section we will further explore potential mechanisms for ketamine's effects but from the perspective of metabolic changes encompassed by changes in protein synthesis and their known relationship to synaptic plasticity: the cellular mechanism considered to be crucial for long-lasting changes in the brain.

Section 3

So far we have described experiments and results pertaining to the electrophysiological effects of ketamine in the hippocampus. In this section we will focus on a series of molecular experiments we performed in order to understand how ketamine impacted *de novo* protein synthesis right after its administration, and whether it had any cell type-specific effects. It is well established that ketamine's primary target is the NMDA receptor, and that this molecular receptor is an essential component of the cellular machinery behind synaptic plasticity (Li et al., 2010; Lodge & Mercier, 2015; Newcomer, 1999; Peltoniemi et al., 2016; Zanos & Gould, 2018; Zhang et al., 2021). Normally, the opening of the channel pore at the center of the NMDA receptor transmembrane domain lets calcium ions flow into the cytoplasm of the cell. These calcium ions serve both to depolarize the adjacent membrane and as a second messenger that initiates complex molecular cascades that influence cellular function. One of these cascades involves activation of calcium-dependent kinases to then control effectors that can increase or decrease synaptic strength via, for example, AMPA receptor insertion/removal, actin cytoskeleton remodeling, receptor phosphorylation,

and *de novo* protein synthesis (Citri & Malenka, 2008; Kang & Schuman, 1996; Kavalali & Monteggia, 2020). Importantly, *de novo* protein synthesis has a central role since without a dynamic mechanism by which the stock of enzymes and effectors is replenished, the remodeling of synaptic architecture would not be able to meet the demands of the fast-paced activity-determined changes neurons experience.

The influx of calcium into the cell triggers specialized enzymes that control the production of proteins. In this way, neurons can dynamically control the energy-intensive process of protein production to meet environmental demands. In a study from 2010, a group of researchers at Yale University from the Duman lab found that ketamine increased the availability of a collection of proteins known to be involved in synaptic plasticity (Arc, GluA1, Synapsin1 and PSD95) *in vitro*, and that this increase was seen to last for several days post ketamine. Additionally they measured increases in the size of dendritic spines. They found that inhibiting the mammalian target of rapamycin (mTOR) eliminated the increase of plasticity related proteins observed after ketamine treatment and that this also decreased new spine growth (Li et al., 2010). Researchers at Vanderbilt University from the Monteggia lab followed up on this study with a special focus on the rapid antidepressant effects of ketamine in mouse models of depression. They found that when ketamine blocked the channel pore of the NMDA receptor and thus inhibited the influx of calcium, the lack of calcium inhibited the eukaryotic elongation factor 2 kinase (eEF2K), which is a calcium-dependent kinase that phosphorylates eEF2. Normally the phosphorylation of eEF2 causes suppression of protein synthesis, but by inhibiting eEF2K, the result of the ketamine-mediated inhibition of NMDA receptors is a desuppression of protein synthesis. They were able to track

how the synthesis of a particularly significant protein for plasticity, brain derived neurotrophic factor (BDNF), was impacted by ketamine's regulation of this pathway (Autry et al., 2011). BDNF's role in synaptic plasticity is well established (Kang & Schuman, 1996; Leal et al., 2015), such that this study was able to rigorously establish *in vivo* at least one synaptic plasticity pathway depending on *de novo* protein synthesis that is influenced by ketamine.

Since the Monteggia study, there have been other pathways that control protein synthesis that have been implicated in the action of ketamine (Adaikkan et al., 2018; Aguilar-Valles et al., 2021). Instead of focusing on particular pathways, we targeted newly synthesized protein products directly by labeling them with fluorescent markers *in vivo*. We used a technique developed by the Schuman lab called fluorescent non-canonical amino acid tagging (FUNCAT), which works by first administering a non-canonical amino acid (in our case a methionine surrogate called azidohomoalanine, AHA) with a free azide group that gets incorporated into nascent proteins and can later be bound to a fluorophore with a free alkyne group through a copper-catalyzed azide-alkyne [3+2] cycloaddition (CuAAC) or 'click' reaction (Dieterich et al., 2010). To be able to label nascent proteins after ketamine treatment in live mice, we injected the AHA retro-orbitally 30 minutes before the first dose of ketamine. The choice of a retro-orbital injection was made based on its ability to reach the brain quickly and incorporate AHA into the nascent proteome before its metabolization, something that has been previously validated by members of the Klann lab (Evans et al., 2025 [preprint]). After injecting AHA retro-orbitally, we proceeded with the same ketamine treatment paradigm performed in the electrophysiology experiments (Figure 2A), so that we could get complimentary

results that shed insight into the molecular nature of ketamine's electrophysiological effects.

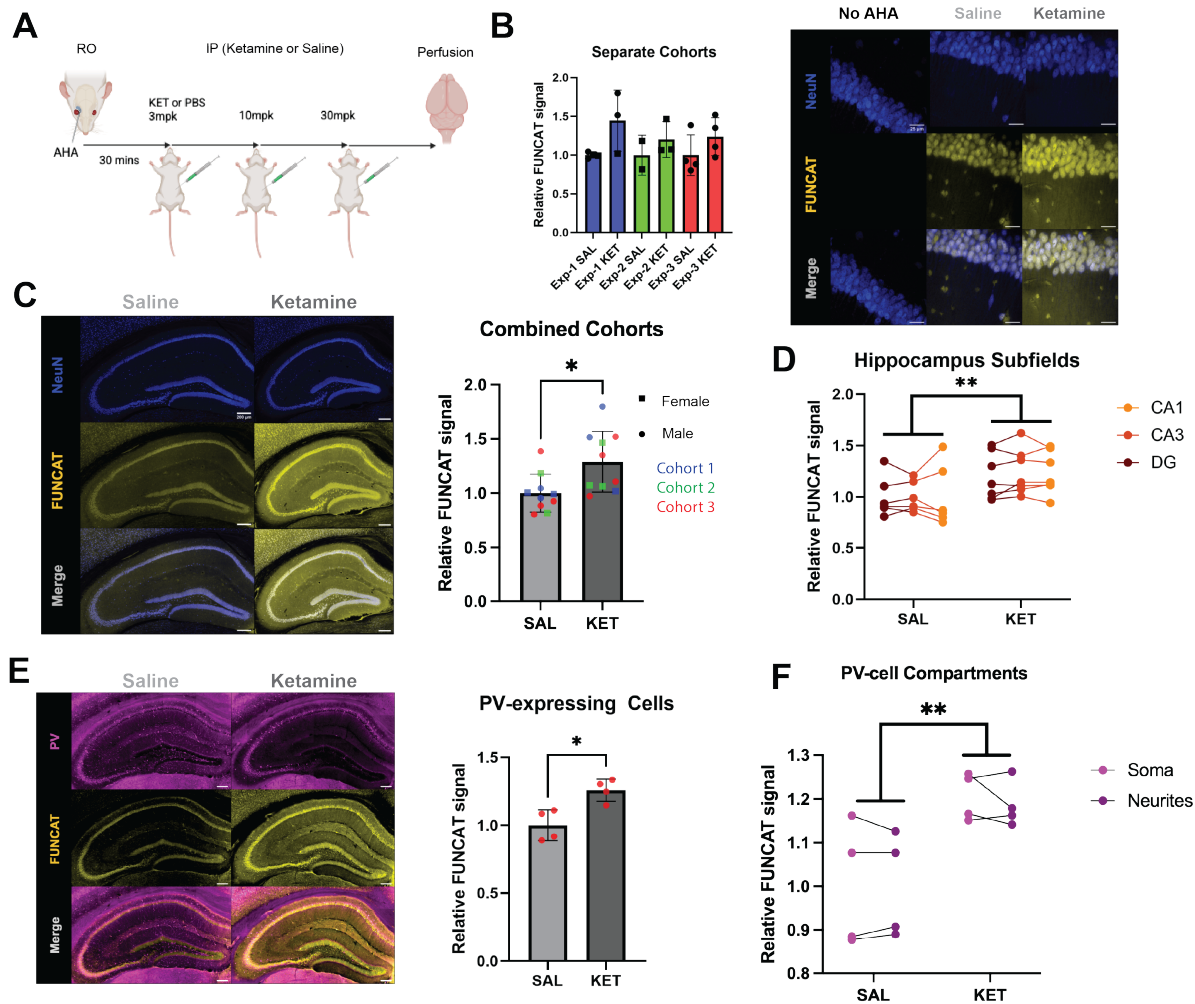


Figure 2. Ketamine treatment increases neuronal protein synthesis in the hippocampus

A Experimental Design. Mice were retro-orbitally (RO) injected *in vivo* with methionine analog azidehomocysteine (AHA) and went through ketamine (or saline control) treatment paradigm consisting of 3 intraperitoneal (IP) injections of increasing dosage [3, 10, 30mg/Kg (mpk)] spaced 30 minutes apart. **B** Hippocampus CA1 neuron histology with FUNCAT signal. Hippocampus tissue with and without AHA labelled proteins was processed *ex vivo* following the FUNCAT protocol, using "click" chemistry (Click-iT Cell Reaction Buffer Kit) with a fluorescent dye (AzDye 647). Neuronal nuclear marker (NeuN) was used to identify neurons and images were acquired using confocal microscope at 40x. AHA labelled samples were quantified for each experimental cohorts, where cohorts 1 and 2 (blue and green) were only given a single dose of ketamine at 15mpk, 30 mins post AHA, while cohort 3 (red) was given all three doses. **C** Quantification of normalized FUNCAT signal in whole hippocampus. Mean pixel intensity was measured from images of hippocampal samples taken using a confocal microscope at 20x to estimate average FUNCAT signal relative to saline control. A significant increase in FUNCAT signal was found in the ketamine treated animals compared to saline controls (unpaired two-tailed t-test $p = 0.0127$). Data come from the three experimental cohorts, each normalized to corresponding control group. Square symbols represent females and circles male. **D** Comparison of FUNCAT signal across hippocampus subfields. Funcat signal relative to saline control was measured at each individual hippocampus subfield (CA1, CA3, DG) and compared to each other. No significant difference was found in signal between subfields but ketamine group showed increased FUNCAT signal across all three subfields (Two-way ANOVA: Drug $p = 0.0032$, Subfield $p = 0.9644$, Interaction $p = 0.9644$). **E** Quantification of FUNCAT signal in PV-expressing cells. Parvalbumin (PV) marker was used to identify PV interneurons from cohort 3. We used Olympus VS200 slide scanner at 20x resolution for the acquisition of these images. A significant increase in FUNCAT signal was observed in ketamine treated PV-cells relative to saline (unpaired two-tailed t-test $p = 0.0103$). **F** Comparison of FUNCAT signal between PV interneuron soma and neurites. PV-cell FUNCAT signal was partitioned between that expressed in the cell bodies and that expressed in the processes. There was not a significant difference in FUNCAT signal between soma and neurites, but both compartments had a significantly higher signal in the ketamine group relative to the saline control (Two-way ANOVA: Drug $p = 0.0021$, Compartment $p = 0.8533$, Interaction $p = 0.8533$).

Post-ketamine treatment (2h post-AHA injection), mice were perfused and the brain was extracted. We sliced at 40 μ m with a Leica vibratome and selected hippocampus slices for the next processing steps. We first performed the fluorescent tagging of our labeled proteins using a Click-iT reaction kit and Alkyne 647 fluorophore. After completing the tagging step, we selected cell-type markers of interest to perform immunolabeling. We used NeuN as a nuclear marker for neurons, as well as parvalbumin (PV) to identify PV interneurons. DAPI was used as a counterstain. After staining, we mounted the tissue and imaged using a Leica confocal microscope at 20x and 40x magnification, as well as a Olympus 200V slide scanner at 20x. Our images showed an increase of protein labeling (FUNCAT signal) in hippocampal neurons of mice treated with ketamine as opposed to those treated with only saline (Figure 2, B and C). No specific subfield within the hippocampus (CA1, CA3, DG) was seen to have a more significant increase in FUNCAT signal (Figure 2D), and identified PV cells also showed an increase in FUNCAT signal for ketamine treated mice when compared to those treated with saline (Figure 2E). Because our hypothesis stated that ketamine is altering particularly inhibitory connections, we compared FUNCAT signal at the soma vs the neurites of PV cells, and found that although there was no significant differences between the two compartments, both had an increase in FUNCAT signal in the ketamine treated group (Figure 2F). Further work needs to be done to better quantify this extra-somatic signal, potentially by using a dendritic marker like MAP2 that allows us to compare FUNCAT signal of excitatory vs. inhibitory processes.

The relative increase in FUNCAT signal for ketamine treated mice is direct evidence that ketamine does in fact potentiate general *de novo* protein synthesis in the

hippocampus *in vivo*. Furthermore, we found that this happens indiscriminately across the different subfields of the hippocampus and that PV-expressing inhibitory neurons also increase their protein synthesis after ketamine, in both their soma and neurites. Increased *de novo* protein synthesis is an essential process that gets activated during synaptic plasticity. Our results further support the hypothesis that ketamine activates molecular mechanisms that can alter synaptic plasticity, and that it can do so rather effectively: within two hours and after a single treatment session. Furthermore, our results relate ketamine's electrophysiological effects with an increase in *de novo* protein synthesis by virtue of the similarity of the two experimental designs. These preliminary results have led us to formulate the following hypothesis. We believe that the disorganization of neural activity due to ketamine, mainly the increase in gamma tuning, requires an increase of protein synthesis. Given the prevalence of changes related to gamma oscillations, we believe that the increase of *de novo* protein synthesis in fast-spiking inhibitory neurons, i.e. PV-expressing cells, are central to these changes. Furthermore, the hypothesized implication of a cell type-specific increase in protein synthesis is that inhibitory neurons will use the newly synthesized proteins to undergo synaptic plasticity and alter the E/I balance of hippocampal networks in a long-lasting manner. At this moment it is not clear exactly in which direction this inhibitory plasticity would tip the scale of the E/I balance, yet perhaps a new series of experiments could elucidate this question. In the subsequent section I will discuss next experiments and open questions for the field and finish with some concluding remarks concerning the implications of the work here presented.

Conclusions and Future Studies

In our investigations into the physiological and molecular effects of ketamine we first confirmed that ketamine alters the spectral power of LFP oscillations, particularly at the theta and gamma range. We then found that ketamine alters the tuning of spiking to gamma oscillations and it can do so to fast-spiking neurons. Then under the same ketamine treatment paradigm we used for the electrophysiological recordings, we found that global protein synthesis, measured as the intensity of FUNCAT signal relative to a saline control, was significantly increased in the cell-dense layers of CA1, CA3 and DG. No significant differences in FUNCAT signal were detected between these subfields of the hippocampus. Furthermore, we immunolabeled PV-expressing interneurons in our hippocampus slices and found that these cells also showed an increase in FUNCAT signal in the ketamine treated animals and did so in both their neurites, the site of E-I and I-I connections, and their somas.

Two of the main hypotheses that have been postulated about ketamine's rapid mechanism of action are first, that ketamine disinhibits neuronal networks in the mPFC by preferentially blocking NMDARs in inhibitory neurons (Gerhard et al., 2020), and second, that ketamine induces rapid synaptic rescaling by blocking NMDARs in the SR of the hippocampus, desuppressing translation of BDNF and triggering BDNF-TrkB signaling, which allows for augmentation of AMPA receptor activation (Autry et al., 2011; Kavalali & Monteggia, 2020). The work presented here has shown initial evidence concerning features associated with synaptic plasticity in hippocampus inhibitory neurons due to ketamine. We showed that the presence of ketamine alters the

electrophysiology of the hippocampus by increasing gamma power and increasing the degree to which neurons are tuned to gamma oscillations. We observed a single case where these effects seemed to persist for two days. These observations must be validated and further researched through future experiments that directly test the long-lasting electrophysiological effects of ketamine and uniquely target fast-spiking inhibitory activity in a more robust way using a larger cohort of mice.

I propose an experimental design where chronic implants that allow for freely moving behavior are used instead of the removable head-fixed preparation used in our experiments. Neural activity in the hippocampus would be recorded before, during and days after ketamine administration. A more granular temporal resolution could also be made by increasing the timepoints of the recordings following Li et al., 2010 where they do 3 to 6 hours and then 1 and 3 days to 1 week. With a cohort of 10 mice per experimental condition (treated vs. control) we could get a clearer, statistically more powerful, picture of the long-lasting effects of ketamine and its variability, for which we have limited preliminary evidence.

Second, we showed that PV-expressing cells have a robust increase in *de novo* protein synthesis. This could mean that PV neurons are undergoing changes such as synaptic plasticity, which requires the new production of proteins. We used gross anatomical segmentation of the hippocampus to isolate the signal from neurites vs. the somas of PV cells, yet we suggest a more rigorous approach that uses MAP2 as a general dendrite marker, such that we can also localize FUNCAT signal to excitatory processes, and imaging at higher resolutions to get clearer signal from these fine structures. We were able to observe a diffuse signal in the SR and SLM of CA1 that

suggests there could be layer specific local protein synthesis, but with the data at hand we can only give a rough estimate at PV-expressing processes. This line of inquiry is exciting since there could layer specific metabolic activity. Layer-specific action could imply that ketamine acts on specific circuits of the hippocampus, such as the tri-synaptic loop. Ketamine could thus be altering the transmission of information from the Schaffer collaterals to the SR associated with internal representations, or from layer III of the MEC to the SLM of CA1, which is associated with current sensorial information (Colgin et al., 2009). Using co-labeling of both PV-expressing cells and MAP2, we could distinguish between inhibitory and excitatory processes and determine if ketamine is increasing the availability of newly synthesized proteins at neural processes in a cell type-specific manner.

Ultimately, our goal is to understand how the electrophysiological effects of ketamine we have observed are related to the changes in protein synthesis we have seen in our data. One question that emerges to target this relationship is whether *de novo* protein synthesis in PV cells is necessary for the increase in gamma power and tuning observed during and potentially after ketamine treatment. I propose a series of experiments that lead up to testing whether the effects on gamma oscillations are observed while protein synthesis is blocked in PV cells. In order to reach this goal, we would first need to validate if our chosen protein synthesis inhibitors (PSIs) do in fact eliminate the increase in FUNCAT signal observed after ketamine treatment. A classic PSI is anisomycin, which although considered a dirty drug that targets biological processes extraneous to protein synthesis, is widely used in the literature (Davis & Squire, 1984).

A more refined approach is to use a method developed by the Klann lab that uses phosphorylation of endogenous eukaryotic initiation factor 2 α (eIF2 α) to rapidly and reversibly inhibit translation. To control phosphorylation of eIF2 α , the method uses transgenic mice that express a modified version of the eIF2 α kinase, inducible protein kinase RNA-activated (iPKR), plus a recombinant protease, nonstructural protein 3/4 (NS3/4), that uniquely targets iPKR for degradation under normal conditions. The drug asunaprevir (ASV) is able to chemically inhibit NS3/4, such that iPKR is free to phosphorylate eIF2 α , which leads to the inhibition protein synthesis (Shrestha et al., 2020). We can use a pan-neuronal promotor to control iPKR and NS3/4 expression in all neurons and compare with results from an experiment that uses anisomycin as a PSI.

The utility and refinement of this method comes into play when we want to block protein synthesis in specific cell types. With the iPKR system we can use a PV-specific promotor to express iPKR and NS3/4 in only PV cells, such that protein synthesis is only blocked in those cells. By using the iPKR system on mice with chronic implants that can record brain activity during ketamine treatment, we can find out if protein synthesis in PV cells is in fact necessary for increased gamma power and tuning due to ketamine. PV cells are also not the only kind of inhibitory neurons that have been implicated in the effects of ketamine, others like somatostatin-expressing interneurons (SST) have also been found to be implicated (Gerhard et al., 2020). By using the iPKR method across selected inhibitory neuron types and additionally collecting cell type-specific proteomic data across our samples, we could get a much more comprehensive understanding of what kinds of cell types are central to ketamine's mechanism of action.

Gamma oscillations have long been interpreted as a neurophysiological signature corresponding to cognitive processing in the brain (Buzsáki & Wang, 2012; Singer, 1999; Uhlhaas et al., 2008). Changes in gamma oscillations have been associated with changes in mental states, which include those states that are seen as pathological in nature such as psychosis. Ketamine rapidly induces changes in gamma oscillations, and does so in a way that is potentially long-lasting (Caixeta et al., 2013; Friesner et al., 2020; Kittelberger et al., 2012). Understanding the molecular neurobiology underlying these physiological changes could shed light on a biological explanation for mental disorders like schizophrenia, where aberrant gamma synchrony is considered a biomarker for the condition (Uhlhaas et al., 2008). Long-lasting changes in gamma oscillations and metabolic changes related to plasticity in fast-spiking inhibitory neurons, particularly in the hippocampus, are only one of many possible frameworks to understand how ketamine-induced changes in psychiatric symptoms could be actualized. This said, it is a compelling model given the recent implications of inhibitory plasticity in learning the statistical structure of an environment (Liao et al., 2024). I would argue that it is this statistical structure of the world which gets altered in mental disorders like schizophrenia and major depression. These psychiatric conditions are precisely considered as pathological because they diverge too far from what is considered a “normal” experience of the world.

Inhibitory drive controls both local and global computation in a neural system, it insures internal parameters that protect the system from over excitation through E-I loops, as well as over inhibition through I-I connections. Understanding the physiological effects of ketamine through a neurobiological lens that integrates neural oscillations and

neural metabolism opens up the door for researchers and clinicians alike to begin approaching research questions and treatment options concerning psychiatric disorders in a more integrated way. By considering physiological biomarkers, like aberrant gamma synchrony, along with a deep understanding of the molecular biology associated to them, such as altered protein synthesis in inhibitory neurons, clinical psychiatry would be able to develop highly targeted remedies to mental disorders like schizophrenia, whose treatment has relied principally on chance pharmacological discoveries with little biological insight and an over reliance on symptoms based medicine.

Methods

Extracellular Recordings During and After Ketamine Treatment: Preliminary results come from analysis done on an existing, previously collected dataset. We used Neuropixels 1.0 electrode probes to record neural activity across the hippocampal CA1-DG axis in head-fixed mice. The mice can explore a small circular arena suspended on an air cushion (Neurotar system), which counteracts locomotion-induced movement. Head-fixing ensures increased stability and reliable electrode localization over extended periods. Extracellular voltage was recorded at 2,500 samples per second for LFP analysis and at 30,000 samples per second for individual spike detection.

Current Source Density (CSD) Analysis: Raw recordings were transformed into LFP by voltage dereferencing and bandpass filtering (1-300Hz). The LFP was then converted into the CSD by applying standard electrostatic analysis using the iCSD toolbox, which estimates the net current flux at neuronal membranes near each electrode.

Electrode Channel Registration to Functional Hippocampal Template: We utilize a recently developed technique in our lab that aligns electrode channels with a functional

hippocampal template based on invariant electric field characteristics (Arend, et al., in prep.). This method allows us to register recordings from different animals (or different sessions within the same animal) to a common anatomical framework, enabling accurate localization of neural activity to the hippocampus.

Gamma Power Spectral Density (PSD) Analysis: We applied fast Fourier transform (FFT) using Welch's method to the CSD signal, isolating gamma oscillations (30-100Hz). The area under the PSD curve was used as a measure of gamma power. Statistical comparisons across doses were performed using repeated-measures (RM) ANOVA with Tukey's multiple comparisons test. Statistical significance was set at $p < 0.05$.

Phase-frequency Tuning of Neural Spiking Analysis: We calculated the degree to which neuronal spiking is organized by neural oscillations to get an estimate of how changes in gamma rhythms impact individual neurons. Using the high-sampling-rate data (30 kHz), we extracted spikes with Kilosort2 and identified spiking units. For each unit, we computed the probability of firing at different phases of oscillations across the frequency spectrum (1-250Hz). This analysis was performed on recordings from animals before ketamine treatment and two days post-treatment to assess whether ketamine alters the coordination of neural firing with gamma oscillations over time.

***In situ* visualization of *de novo* protein synthesis using FUNCAT *in vivo*:** We used Fluorescent Non-Canonical Amino Acid Tagging (FUNCAT) to visualize *de novo* protein synthesis after ketamine treatment. This method employs L-Azidohomoalanine (AHA; Vector Labs Ref:CCT-1066-100, Lot:5543), a methionine analog bearing a free azide group, which is incorporated into nascent proteins by the cell's native translational machinery. Following AHA incorporation, a fluorescent dye (AzDye 647; Vector Labs Ref:CCT-1301-1, Lot:32722) is attached via copper-catalyzed azide-alkyne [3+2] cycloaddition ("click" chemistry), enabling detection of newly synthesized proteins (Dieterich et al., 2010). We use the Click-iT Cell Reaction Buffer Kit (Invitrogen Ref:C10269, Lot:2816109) for this reaction. AHA was delivered via retro-orbital injection

into awake mice. AHA incorporation occurs optimally over a two-hour window following injection (Evans et al., 2025 [preprint]). After this period, mice were perfused, and brains extracted and fixed in 4% paraformaldehyde (PFA). Post-fixation, fluorescent tagging is performed on hippocampal slices and prepared for imaging.

Ketamine treatment paradigm: We followed the same treatment paradigm that was used in the electrophysiology experiments that showed increases in gamma power and tuning, where dose-dependent analysis of ketamine's effects was also a priority. Mice received a 10mg/mL ketamine solution (Ketaset RS ketamine hydrochloride, diluted in sterile saline) administered intraperitoneally (IP) at three escalating doses based on previous studies in the lab: 3mg/kg, 10mg/kg, and 30mg/kg, spaced 30 minutes apart, which is known to be the extent of time for the rapid effects of ketamine (Autry et al., 2011). The first ketamine injection occurs 30 minutes after the AHA injection. Mice were allowed to recover in their home cages between injections. Following the final 30-minute period, mice were perfused with 1X PBS followed by 4% PFA.

Immunofluorescent labeling of inhibitory neurons: Fixed brains were sliced at 40µm thickness using a Leica vibratome. Slices were blocked and permeabilized in 5% normal goat serum and 0.3% Triton X-100 in PBS. After PBS washes, slices got processed using the Click-iT Reaction Kit. To identify interneuron subtypes, slices were incubated with primary antibody against parvalbumin (MSxParvalbumin; Millipore MAB1572 Lot:2961807) followed by goat anti-mouse Alexa Fluor 488 secondary antibody. Nuclei were counterstained with DAPI. Sections were mounted on frosted slides using 60% glycerol as the mounting medium. We also used a neuronal nuclear protein (NeuN) primary antibody (Synaptics Systems 266-014, Lot:1-6) as a general neuronal marker, which together with DAPI was used to identify the different regions of the hippocampus through known anatomical landmarks in the background that can be validated using a mouse brain atlas.

Image acquisition and analysis: Images were acquired using a Leica SP8 confocal microscope (20x and 40x objectives) and an Olympus VS200 slide scanner (20x

objective) for whole hippocampal sections. Image analysis was performed using custom ImageJ macros and complementary Python-based image processing pipelines I developed. Analyses include sampling defined regions of interest (ROI) based on natural differences in the signal background, as well as cell segmenting, and quantifying mean pixel intensity within each ROI. Statistical comparisons between experimental groups were performed using unpaired two-tailed Student's t-tests or 2-way ANOVA when comparing more than two groups. Statistical significance was set at $p < 0.05$. Analysis code (ImageJ macros and Python notebooks) is available upon request.

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